Yunkang Cheng | 程允康

m13700702234@163.com | https://cyk579.github.io/
Department of Automation, Tsinghua University, Beijing, China

EDUCATION

Tsinghua University, Beijing, China **2024.8 - 2029 (Expected)**

Bachelor of Engineering in Automation, Department of Automation

- GPA: 3.7/4.0.
- Selected coursework: Linear Algebra (A-), Calculus A(1) (A-), Object-Oriented Programming Practicum (A), Digital Electronics Laboratory (A-), Electronic Technology Course Design (A-).

RESEARCH EXPERIENCES

Humanoid Roller-Skating with Adversarial Motion Priors **2025.12 - Present**

Project Lead, Robot Control Laboratory, Tsinghua University | Advisor: Prof. Mingguo Zhao

Learning Roller-Skating Motions of Humanoid Robots Based on Adversarial Motion Priors

Yunkang Cheng*, *Yutong Wu**, *Menghan Li, Shihe Zhou, and Mingguo Zhao*

CLAWAR 2026, oral presentation. * Equal contribution; listed first.

arXiv:2607.10815 | Project website

- Lead a research project on passive-wheel humanoid skating; develop motion-control algorithms and training networks, and coordinate project progress and experimental work.
- Develop AMP-PPO policies for Pump Glide and Push Glide using retargeted human motion-capture data, gait-specific rewards, and domain randomization in Isaac Lab.
- Contribute to an integrated simulation-to-hardware pipeline on Booster T1, including wheel-contact modeling, MuJoCo cross-simulation evaluation, and real-robot trials. The policies control 21 body joints at 50 Hz; skate wheels remain passive.
- Evaluate gait rhythm, velocity response, posture, and support timing in simulation; demonstrate both skating gaits on hardware. Present the work at CLAWAR 2026 in Cranfield, UK.
- Ongoing work: motion-imitation policies for skating starts, contact-model calibration, and transitions between starting and gliding. The project is supported by the national undergraduate innovation program and Tsinghua's Student Academic Research Advancement Program (Class B).

Humanoid Robot Soccer, Tsinghua Hephaestus **2025 - 2026**

Core Team Member | Advisor: Prof. Mingguo Zhao

- Develop high-speed obstacle-avoidance algorithms for humanoid soccer, including navigation for returning to the field and reaching assigned positions in dynamic match environments.
- Integrate and debug navigation behavior with teammates during competition preparation and on-site matches; contribute to the team's RoboCup and World Humanoid Robot Games campaigns.

SKILLS AND INTERESTS

Programming: Python, C/C++, MATLAB.

Tools and platforms: PyTorch, Isaac Lab, Isaac Gym, MuJoCo; Booster T1 and Qmini.

Research interests: Robot learning, reinforcement and imitation learning, humanoid locomotion, contact dynamics, and sim-to-real transfer.

SELECTED ENGINEERING PROJECTS

Passive Walking Dynamics and Stability Analysis **2026.10**

Humanoid Robotics Course Project, Tsinghua University

- Model a passive biped with curved feet in MATLAB, combining continuous dynamics with impact resets; analyze the effects of mass, center-of-mass placement, and spring stiffness on walking.
- Compute fixed points of the step-to-step return map and assess local stability via its spectral radius; validate 48 parameter settings with 50-step perturbation simulations.

Qmini Bipedal Robot Assembly and Modification **2025 Fall**

Experience of Manufacturing Engineering Course Project, Tsinghua University

- Build, modify, and debug a Qmini bipedal robot, gaining practical experience in mechanical assembly, hardware integration, and motion-control testing.

HONOURS AND AWARDS

Competition results below are team awards.

RoboCup, Humanoid Soccer League - Large Division **2026**

Champion, Tsinghua Hephaestus; Incheon, Republic of Korea.

World Humanoid Robot Games, 5v5 Soccer **2025, 2026**

Champion, Tsinghua Hephaestus; 2026 category: large robots.

RoboCup China Open, Humanoid Large League **2026**

First Prize (Champion), and Champion of the Technical Challenge.

Tsinghua Future Robotics Challenge **2026**

First Prize; Team Captain.

Second Embodied Intelligence Challenge, Department of Automation **2025**

First Prize; Deputy Team Captain.

National Encouragement Scholarship, Tsinghua University **2026**

Also awarded the Comprehensive Excellence Award.

Spark Fellowship, Tsinghua University **2026**

Selected for the 20th cohort of the university's student innovation talent program.

Tang Chung Ying Scholarship, Tsinghua University **2024**

EXTRACURRICULAR EXPERIENCE

Tang Chung Ying Love Society, Tsinghua University **2026.2 - Present**

Vice President for General Affairs (since 2026.5); previously Head of General Affairs

- Coordinate general affairs, financial reporting, monthly reports, and liaison with the Tang Foundation; organize departmental responsibilities and volunteer recruitment.

Embodied Intelligence Industry Field Study, Guangdong **2026.8**

Team Leader, Department of Automation

- Lead a 17-member team on an eight-day study in Shenzhen and Guangzhou, covering embodied intelligence, drones, and autonomous driving; coordinate preparation, interviews, and reporting. Awarded the department's Silver Prize for social practice.

Rural Revitalization Fieldwork, Fangzheng, Heilongjiang **2026.1**

Team Leader, Tsinghua University

- Lead a 15-member team and organize robotics outreach in schools. Help establish the Shuimu Yike online teaching program at Fangzheng No. 4 Primary School; the team received a Gold Award from the School of Architecture and a university-level Silver Award.